

Stellenbosch University
Department of Industrial Engineering
Operations Research 415
Tutorial 4: 14 April 2026 (on §17.1–§17.6)

- (1) Suppose every student in Stellenbosch subscribes to one of three music streaming services: Echo, Pulse, or Nova, renewing the subscription every month. Each month 17% of all Echo users switch to Pulse, while 15% of all Echo users switch to Nova. Furthermore, 11% of all Pulse users switch to Echo, while 12% of all Pulse users switch to Nova each month. Finally, 14% of all Nova users switch to Echo, while 9% of all Nova users switch to Pulse each month.
- (a) Model the transition in streaming service subscriptions among students in Stellenbosch as a Markov chain. Produce both the transition probability matrix and a graphical representation of the chain. Is the chain ergodic? Motivate.
 - (b) If a student now uses Echo, what is the probability that the student will still be using Echo two months from now?
 - (c) If a student now uses Echo, what is the probability that the student will be using Nova two months from now?
 - (d) If a student now uses Pulse, what is the probability that the student will be using Echo three months from now?
 - (e) If 35% of all students now use Echo, and 25% of all students now use Pulse, what percentage of students will be using Nova two months from now?
 - (f) Which streaming service will achieve the largest market share among students in Stellenbosch in the long run? Motivate your answer.
- (2) The Department of Industrial Engineering has two high-capacity label printers. During any day, each printer that is in working condition at the beginning of the day has an 18% chance of breaking down during that day. If a printer breaks down, a technician is called out and arrives at the start of the next day. The technician then spends two full days repairing the printer, after which it is back in working condition. For example, if a printer breaks down during day 6, it will be back in working condition at the beginning of day 9.
- (a) Produce the transition matrix of an appropriate Markov chain model for the number of label printers in operation at the beginning of each day. (Hint: There should be six states in your chain.)
 - (b) Is the chain ergodic? Motivate.
 - (c) Given that both printers are in working condition at the start of day x , what is the probability that exactly one printer will be in working condition at the start of day $x + 1$?
 - (d) Given that both printers are in working condition at the start of day x , what is the probability that exactly one printer will be in working condition at the start of day $x + 2$?
 - (e) Given that both printers are in working condition at the start of day x , what is the probability that both printers will be broken down at the end of day $x + 2$?

- (f) Given that both printers are in working condition at the start of day x and that one printer breaks down during day x , show that it is impossible that both printers will be returned to working condition by the technician at the start of day $x + 5$.
- (3) Three autonomous defence drones enter a three-way duel. Drone A has probability $\frac{5}{8}$ of destroying the drone at which it fires, drone B has probability $\frac{1}{2}$ of destroying the drone at which it fires, and drone C has probability $\frac{1}{4}$ of destroying the drone at which it fires. The drones fire simultaneously in successive duel rounds, with each drone firing at the strongest remaining opponent.
- Explain why the situation can never occur where drones A and B have survived, but not drone C .
 - Model the duel as a Markov chain, taking as states of the chain the subsets of $\{A, B, C\}$ representing the surviving drones. Present the transition probability matrix of the chain.
 - Is the Markov chain in (b) above ergodic, absorbing, or neither? Motivate.
 - For how many duel rounds do you expect all three drones to survive?
 - What is the probability that all three drones will survive at least two rounds?
 - What is the probability that all three drones will survive exactly two rounds?
 - What is the probability that exactly two drones will survive after the first round?
 - What is the probability that exactly one drone will survive after the first round?
 - How many rounds do you expect drone A to survive before the duel ends?
 - How many rounds do you expect drone C to survive before the duel ends?
 - What is the probability that the duel ends with exactly one drone surviving?
 - How many rounds do you expect the duel to last?
- (4) A student enrolls for a three-year analytics diploma programme. Each year there is a probability 0.12 of the student being excluded from the programme, a probability 0.18 of failing and having to repeat the year, and a probability 0.70 of passing and moving on to the next year of study. In the fourth year, moving on means graduating.
- Model the study progression of the student as a Markov chain. (Hint: There should be five states in your chain.)
 - Is the Markov chain in (a) above ergodic, absorbing, or neither? Motivate.
 - How long should the student expect to spend in the second year of study?
 - How long should the student expect to be enrolled for the programme?
 - What is the probability that the student will graduate?
 - What is the probability that the student will graduate, given that the student passes the second year of study?